CHAPTER 12

TIME SERIES ANALYSIS AND INDEX NUMBERS

CHAPTER ESSENCE AT A GLANCE:
Master time series components (Secular trend, Seasonal, Cyclical, Irregular), Method of Moving Averages, Least Squares linear trend fitting, and weighted price index numbers (Laspeyres, Paasche, Fisher).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Components of Time Series:** Time series data is decomposed into four movements: Secular Trend T (long-term direction), Seasonal Variation S (periodic within a year), Cyclical Fluctuation C (multi-year wave), and Irregular Movement I (random shocks).

• **Additive vs Multiplicative Models:** Additive Model: Y = T + S + C + I (assumes components are independent in absolute units); Multiplicative Model: Y = T * S * C * I (assumes percentage geometric compounding).

• **Trend Estimation Methods:** Moving Averages smooth out short-term fluctuations by averaging overlapping windows; Method of Least Squares fits mathematical trend line Y = a + bX with normal equations.

• **Weighted Index Numbers and Tests:** Laspeyres uses base year quantities; Paasche uses current year quantities; Fisher's Ideal Index is geometric mean of Laspeyres and Paasche, satisfying Time Reversal and Factor Reversal tests.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$Y = a + bX \Longleftrightarrow \sum Y = na + b\sum X \quad \text{and} \quad \sum XY = a\sum X + b\sum X^2$$

Normal Equations for Linear Least Squares Secular Trend

$$P_{01}^{L} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 \quad \text{and} \quad P_{01}^{P} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$$

Laspeyres Base-Weighted and Paasche Current-Weighted Indices

$$P_{01}^{F} = \sqrt{P_{01}^{L} \times P_{01}^{P}} \rightarrow \text{Fisher Ideal Index}$$

Geometric Mean Formulation Satisfying Both Reversal Tests

$$\text{Reversal Criteria} \Longleftrightarrow P_{01} \times P_{10} = 1 \quad \text{and} \quad P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

Time Reversal and Factor Reversal Test Identities

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Laspeyres index tends to overestimate inflation due to base year quantity weights (ignores consumer substitution); Paasche index tends to underestimate inflation.

• **Alert:** Fisher's Ideal Index satisfies both Time Reversal Test and Factor Reversal Test, which is why it is uniquely termed 'Ideal'.

• **Alert:** When fitting a linear trend using Least Squares with odd number of years, set origin X = 0 at the middle year so sum X = 0, simplifying to a = (sum Y) divided by n and b = (sum XY) divided by (sum X^2).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the four components of a Time Series with suitable practical examples. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Secular Trend (T):** The smooth, long-term general tendency of data to increase or decrease over a prolonged period. Example: Long-term increase in India's agricultural output due to improved technology. **[1 Mark]**

• **2. Seasonal Variations (S):** Periodic, regular, and repetitive fluctuations occurring within a time span of one year caused by climate, seasons, or social customs. Example: Surge in ice-cream sales in summer or umbrella sales in monsoon. **[1 Mark]**

• **3. Cyclical Fluctuations (C):** Wave-like oscillatory movements recurring over longer durations (3 to 10 years) following business prosperity, recession, depression, and recovery phases. **[1 Mark]**

• **4. Irregular (Random) Variations (I):** Erratic, unscheduled shocks caused by unforeseen catastrophes like earthquakes, wars, floods, and pandemics. **[1 Mark]**

2. Fit a straight line trend by the method of least squares to the following production data: Year: 2018 (Production=10), 2019 (12), 2020 (14), 2021 (16), 2022 (18). Estimate production for the year 2024. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Origin Shift:** n = 5 (odd). Let middle year 2020 be origin (X = Year - 2020). X takes values: -2, -1, 0, +1, +2. sum X = 0. **[1 Mark]**

• **Calculations:** sum Y = 10 + 12 + 14 + 16 + 18 = 70. sum X^2 = $(-2)^2$ + $(-1)^2$ + 0 + 1^2 + 2^2 = 10. sum XY = (-2 × 10) + (-1 × 12) + (0 × 14) + (1 × 16) + (2 × 18) = -20 - 12 + 0 + 16 + 36 = 20. **[1 Mark]**

• **Coefficients:** a = sum Y divided by n = 70 divided by 5 = 14. b = sum XY divided by sum X^2 = 20 divided by 10 = 2. Trend Line: Y = 14 + 2X. **[1 Mark]**

• **Estimate for 2024:** For 2024, X = 2024 - 2020 = 4. Y = 14 + 2 * 4 = 14 + 8 = 22 units. **[1 Mark]**

3. If Laspeyres Price Index is 120 and Paasche Price Index is 125, calculate: (a) Fisher's Ideal Index, (b) State which tests are satisfied by Fisher's Index. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **(a) Fisher Ideal Index:** Formula: P_{01}^{F} = square root of [Laspeyres * Paasche] = square root of [120 × 125] = square root of 15,000 = 122.47. **[2 Marks]**

• **(b) Tests Satisfied:** Fisher's Ideal Index satisfies: (1) Time Reversal Test: P01 * P10 = 1; (2) Factor Reversal Test: P01 * Q01 = Value Ratio. **[2 Marks]**

4. Explain the Method of Moving Averages for measuring trend. State one advantage and one limitation. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Method:** Successive arithmetic means are calculated for overlapping blocks of k consecutive time periods, smoothing out seasonal and cyclical fluctuations. **[1 Mark]**

• **Advantage:** Extremely simple, objective, and does not assume any rigid mathematical curve for the trend. **[1 Mark]**

• **Limitation:** Loses trend values at the beginning and end of the series, and cannot be used directly for future forecasting. **[1 Mark]**

5. Distinguish between Additive Model and Multiplicative Model of time series decomposition. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Additive Model:** Y = T + S + C + I. Assumes the four components are independent and expressed in original absolute units. **[1 Mark]**

• **Multiplicative Model:** Y = T * S * C * I. Assumes components are interdependent and seasonal, cyclical, and irregular effects operate as percentage multipliers of trend. **[1 Mark]**

6. State the mathematical condition for the Time Reversal Test in index numbers. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Mathematical Condition:** P01 * P10 = 1 (ignoring the factor 100). The index calculated forward in time from base year 0 to current year 1 multiplied by the index calculated backward from year 1 to year 0 must equal unity. **[2 Marks]**